Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

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Project Proposal Report

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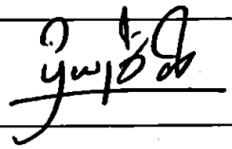
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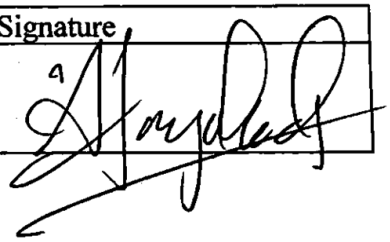
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Training autonomous vehicles depends on large amounts of labelled sensor data covering object detection, movement prediction, and navigation. Real driving footage almost always captures faces, license plates, and repeated location patterns that can identify people, so using it directly for AI training runs into serious problems under laws like the GDPR. Developers who step away from real data then rely on synthetic datasets, but none of the tools available today can back those datasets with any formal privacy proof.

Simulation platforms like CARLA and AirSim reduce how much real data is needed, but they are still calibrated against real driving records, and that calibration step is where the problem reappears. A generative model trained in real data can retain sensitive patterns, leaving synthetic outputs open to membership inference and re-identification attacks. No existing tool enforces privacy during the generation process itself. This work integrates epsilon-differential privacy (ε-DP) at both the calibration and training stages. The Laplace mechanism will protect aggregate calibration statistics before they reach the simulator, and per-sample gradient clipping with Gaussian noise injection will be enforced during training through the Opacus library, with a Renyi Differential Privacy accountant tracking the total budget throughout.

The outcome should be a repeatable generation workflow where privacy guarantees are measured rather than assumed, validated through membership inference attack tests against the final synthetic dataset. On the social side, fewer people captured on public roads will be exposed through AI training pipelines, which matters for public trust in autonomous systems. Commercially, AV developers get a legally defensible path to scalable training data without the overhead of data licensing or compliance review. The project connects to SDG 9 (innovation), SDG 11 (sustainable transport), and SDG 16 (responsible data governance).

Keywords: *Differential Privacy, Synthetic Data Generation, Autonomous Vehicles, Membership Inference Attacks, Privacy-by-Design*

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I would like to thank Prof. Anuradha Jayakody for supervising this project. His feedback was direct and useful, and his input during the early stages helped me narrow down the scope of the privacy integration work into something realistic and testable within the project timeline.

Working through the group project with IT22541284, IT22110220, and IT22066916 has shaped how I approached this component. Talking through where the individual parts connect, particularly how the privacy constraints interact with the simulation environment and the attack testing framework, pushed me to think more carefully about the design decisions described in this proposal.

I am also grateful to the Department of Information Technology at SLIIT for the access to resources and the academic environment that made this kind of research possible at undergraduate level.

TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 7
LIST OF FIGURES 8
LIST OF ABBREVIATIONS 9
LIST OF APPENDICES 10
1. INTRODUCTION 11
1.1 Background and Context 11
1.2 Literature Review 12
1.3 Research Gap 13
2. OBJECTIVE 15
2.1 Main Objective 15
2.2 Specific Objectives 15
3. METHODOLOGY 16
3.1 Research Approach 16
3.2 Selecting the Privacy Mechanism and Library 16
3.3 Privacy-Aware Calibration 17
3.4 Differentially Private Training of the Generative Model 17
3.5 Managing the Privacy Budget 17
3.6 Validation Through Attack Testing 18
3.7 Ethical Considerations 18
4. HIGH-LEVEL SYSTEM ARCHITECTURE 19
4.1 Overview 19
4.2 Sub-components 19
4.3 Data Flow 20
4.4 Security and Scalability 20
5. USER REQUIREMENTS 22
5.1 Stakeholders 22
5.2 Functional Requirements 22

5.3 Non-Functional Requirements 22
6. COMMERCIALIZATION PLAN 24
6.1 Target Market and Customer Persona 24
6.2 Value Proposition 24
6.3 Revenue Model 24
6.4 Cost Estimation 24
6.5 Pricing Strategy 24
6.6 Competitive Advantage 24
6.7 Intellectual Property Considerations 24
7. BUDGET AND JUSTIFICATION 25
8. WORK BREAKDOWN STRUCTURE (WBS) 26
8.1 Task Breakdown per Member 26
8.2 Timeline Alignment 27
8.3 Balanced Workload Distribution 27
9. GANTT CHART 28
10. REFERENCES LIST 29
11. APPENDICES 30
Appendix A: Differential Privacy Mechanism Summary 30
Appendix B: Risk Analysis 30
Appendix C: Detailed Diagrams 31
Appendix D: Ethical Documents 31

LIST OF TABLES

Table 7-1: The Budget 25
Table 8-1: Task Breakdown per Member 26
Table 8-2: Timeline Alignment 27
Table 8-3: Balanced Workload Distribution 27
Table 9-1: Gantt Chart 28

LIST OF FIGURES

Figure 4-1: High-Level System Architecture 19
Figure 4-2: Data Flow Diagram 20

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
AV	Autonomous Vehicle
AWS	Amazon Web Services
CARLA	Car Learning to Act (open-source AV simulator)
CCPA	California Consumer Privacy Act
CGAN	Conditional Generative Adversarial Network
CPU	Central Processing Unit
CSV	Comma-Separated Values
DL	Deep Learning
DP	Differential Privacy
FID	Fréchet Inception Distance
FR	Functional Requirement
GAN	Generative Adversarial Network
GDPR	General Data Protection Regulation
GB	Gigabyte
GPU	Graphics Processing Unit
IP	Intellectual Property
IT	Information Technology
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupee
mAP	Mean Average Precision
MIA	Membership Inference Attack
ML	Machine Learning
NFR	Non-Functional Requirement
NVIDIA	NVIDIA Corporation (GPU and simulation platform vendor)
PDPA	Personal Data Protection Act (Sri Lanka, No. 9 of 2022)
RDP	Renyi Differential Privacy
ROC	Receiver Operating Characteristic
SaaS	Software as a Service
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
TF	TensorFlow
UI	User Interface
USD	United States Dollar
VAE	Variational Autoencoder
ε	Epsilon (privacy budget parameter in differential privacy)

LIST OF APPENDICES

Appendix A: Differential Privacy Mechanism Summary 30
Appendix B: Risk Analysis 30
Appendix C: Detailed Diagram 31
Appendix D: Ethical Documents 31

1. INTRODUCTION

1.1 Background and Context

Self-driving vehicles use machine learning at every stage of the driving task. The perception system has to detect and classify objects in real time. The prediction system has to anticipate how pedestrians and other vehicles will move. The planning system has to turn those predictions into steering, acceleration, and braking decisions. Each of these models requires training data, and the amounts involved are large. Waymo reported in 2023 that its fleet had driven over 20 million miles on public roads to generate training data, and researchers at Carnegie Mellon University estimated that a production-grade AV perception model may require upwards of 10 million labelled sensor frames before reaching deployment-level accuracy. The global autonomous vehicle market was valued at approximately USD 54 billion in 2023 and is projected to reach USD 556 billion by 2026 according to Allied Market Research [12], which means the data problem will only grow as more companies enter the space.

The data those vehicles collect is not neutral. Dashcams and roof-mounted cameras record human faces in detail. LiDAR scanners capture the locations of pedestrians and other vehicles with high spatial precision. When the same routes are driven repeatedly, the resulting datasets can reveal where people live, where they work, and how their daily routines are structured. Under the GDPR, this type of data qualifies as personal data subject to strict controls [12]. The European Data Protection Board reported in 2022 that GDPR fines across member states exceeded EUR 1.6 billion in that year alone, reflecting how seriously regulators take violations. Article 5 of the regulation requires that data collected for one purpose not be freely reused for another, which creates a direct legal barrier for AV developers who want to repurpose road test footage for AI model training. In the Sri Lankan context, the Personal Data Protection Act No. 9 of 2022 introduces similar obligations, meaning this is not only a European regulatory concern.

The response from the research community has been to generate synthetic training data using simulation environments. Platforms like CARLA, AirSim, and NVIDIA Omniverse can produce large volumes of photo-realistic labelled sensor data without placing cameras on real roads. But these platforms have a fundamental limitation: to make their outputs statistically realistic enough to be useful, they typically need to be calibrated against real-world data. That calibration step is where the privacy risk re-enters the picture. A generative model that has been trained or calibrated on real driving footage can retain statistical traces of the people and places it saw, and those traces can sometimes be extracted from the model or from its outputs.

The affected parties extend well beyond the companies building these systems. Pedestrians and drivers whose data was captured during road testing have a reasonable expectation that it will not end up in an AI training pipeline. Regulators need to be able to verify that synthetic datasets do not circumvent the protections that data protection law is supposed to provide. Researchers need training data that is shareable without legal risk. At the moment, none of these needs are well served by existing tools.

This research sits within the AIMS (Autonomous Intelligent Machines and Systems) cluster at SLIIT and falls under the Cyber Security specialization. Its primary concern is privacy-by-design in AI data pipelines. It connects to SDG 9 through the goal of making AI infrastructure more secure, to SDG 11 through safer and more trustworthy autonomous transport, and to SDG 16 through demonstrating that data governance in AI development can be made accountable and auditable.

1.2 Literature Review

Differential privacy was formally defined by Cynthia Dwork in 2006 as a mathematical framework for limiting what an observer can learn about any individual from the output of a computation [1]. The core guarantee is quantified by a parameter called epsilon. A lower epsilon value means stronger privacy protection, but it also means more noise is added to the computation, which can reduce accuracy. The challenge for any application of differential privacy is choosing an epsilon value that provides meaningful protection without making the output useless.

Applying differential privacy to deep learning was demonstrated at scale by Abadi and colleagues in 2016 [2]. Their approach clips the gradient contribution of each individual training sample before aggregating across the batch, then adds calibrated Gaussian noise to the aggregated gradient. This prevents any single training record from having too large an influence on the model weights, which is the mechanism through which memorization occurs. The privacy cost of each training step is tracked using an accounting method called the moments accountant, and the cumulative cost across all training steps gives the total epsilon spent. Their implementation became the foundation for TensorFlow Privacy.

Yousefpour and colleagues extended this work by building Opacus, a PyTorch library that makes differentially private training accessible without requiring changes to model architecture [3]. Opacus handles per-sample gradient computation automatically, which was previously a significant engineering obstacle. It also uses a more precise privacy accounting method called Renyi Differential Privacy (RDP), which gives tighter bounds on the total epsilon than the original moments accountant. Opacus is now widely used in privacy-sensitive machine learning research and is the most practical tool available for the kind of work this project requires.

On the AV data side, Dosovitskiy and colleagues presented CARLA as an open-source urban driving simulator capable of generating synchronized multi-sensor data with automatic labelling [4]. CARLA can produce camera images, LiDAR point clouds, and radar returns alongside ground-truth bounding boxes and semantic segmentation masks. AirSim, developed by Microsoft, offers similar capabilities with a stronger focus on photorealistic rendering [5]. Neither platform addresses the privacy question. They treat data generation as a purely technical problem of realism and variety, with no consideration of what information might be retained from real-world calibration data.

Shokri and colleagues demonstrated in 2017 that machine learning models are vulnerable to membership inference attacks, where an adversary tries to determine whether a specific record was in the training set [6]. The attack works by observing how confidently the model predicts the class of a given input. Models tend to be more confident on records they were trained on, and this difference is measurable. This is directly relevant to AV synthetic data: if a generative model was calibrated on real driving data and an attacker has access to that model or its outputs, they may be able to determine which individuals or locations were in the calibration set.

Carlini and colleagues extended this concern by showing that large generative models can directly memorize training examples and reproduce them verbatim under certain conditions [7]. This means the risk is not just statistical inference but literal data extraction. Narayanan and Shmatikov showed in 2008 that even datasets that appear anonymized can be re-identified by cross-referencing with external data sources, a technique they demonstrated on the Netflix

Prize dataset [8]. These results together make the case that informal privacy measures like face blurring or plate obscuring are insufficient.

Research on differentially private GANs has shown that it is technically feasible to combine generative models with formal privacy guarantees. Generative adversarial networks, introduced by Goodfellow and colleagues [11], form the basis for this line of work. Xie and colleagues proposed DP-GAN, which applies differentially private stochastic gradient descent during discriminator training [9]. Torkzadehmahani and colleagues proposed DP-CGAN with label conditioning and a refined accounting strategy [10]. Both papers demonstrate the concept on standard image benchmarks like MNIST. Neither applies the approach to multi-modal AV sensor data, which involves a much more complex combination of data types and poses different challenges for privacy accounting.

The gap this project fills is the absence of a formal privacy integration method designed specifically for the AV synthetic data generation workflow. No existing work has applied epsilon-DP to the calibration and generation pipeline for multi-modal AV datasets, validated the results against domain-relevant attack tests, or produced a documented methodology that others in the field could reproduce.

1.3 Research Gap

Simulation platforms for AV training have become technically capable, but the question of what information they retain from real-world calibration data has not been studied with any rigor. Dosovitskiy et al. [4] and Shah et al. [5] both present their simulators primarily as realism and scalability tools. Neither paper mentions privacy constraints on the calibration process, and neither evaluates whether outputs from a calibrated simulator carry statistical traces of the real data used to configure it. When a simulator is configured to match the traffic patterns or pedestrian behavior of a specific city, it uses statistical summaries derived from real observations, and those summaries are currently computed without any privacy protection.

Differentially private training techniques exist for neural networks, but the work by Abadi et al. [2] and Yousefpour et al. [3] was developed and validated on single-modality classification tasks using datasets like MNIST and CIFAR-10. AV sensor data is fundamentally different. It is multi-modal, combining camera images with LiDAR point clouds and radar returns. It is spatially correlated, with the same pedestrian appearing across many frames from different angles. It is temporally structured, with movement trajectories that carry more identifying information than any single frame. No published work has adapted these privacy accounting methods to the multi-modal, temporally structured nature of AV training data.

There is also no established way to evaluate privacy in synthetic AV datasets. The membership inference attack framework developed by Shokri et al. [6] and the data extraction techniques demonstrated by Carlini et al. [7] were designed for classification and language models. Neither account for the spatial and temporal structure of driving sequences. Xie et al. [9] and Torkzadehmahani et al. [10] validated their differentially private GAN approaches against standard image benchmarks, not against multi-sensor AV outputs. There is no agreed threshold for an acceptable epsilon value in this domain, and no standard protocol for testing re-identification risk in synthetic driving trajectories.

Taken together, these three gaps mean that no existing study has treated privacy as a design constraint on the AV data generation pipeline itself, validating the outcome against domain-specific attacks, or produced a methodology that others could replicate. The work by Xie et al.

[9] and Torkzadehmahani et al. [10] comes closest, but neither extends to multi-modal sensor data nor covers the calibration step. This project fills all three gaps by building privacy enforcement into the generation process from the start, using tools already proven in deep learning research, and evaluating the results with attack tests run directly on synthetic AV data.

2. OBJECTIVE

2.1 Main Objective

To integrate epsilon-differential privacy (ε-DP) into the synthetic autonomous vehicle data generation pipeline, such that the cumulative privacy budget does not exceed ε = 10, membership inference attack success rates remain at or below 55%, and the resulting synthetic dataset sustains AV perception model accuracy within 15% of the real-data baseline.

2.2 Specific Objectives

- To select a differentially private training library by evaluating Opacus and TensorFlow Privacy against criteria of per-sample gradient support, RDP accounting compatibility, and generative model integration.
- To enforce ε-DP during generative model training through per-sample gradient clipping and Gaussian noise injection, verified across at least three ablation configurations with varying clipping norms and noise multipliers.
- To protect real-world calibration inputs by applying the Laplace mechanism to aggregate statistical queries prior to simulator use, constraining the calibration step's ε contribution to below 20% of the total privacy budget.
- To enforce the cumulative privacy budget ceiling of ε = 10 using the Renyi Differential Privacy accountant, with automatic training termination upon reaching the configured limit and a full budget log retained per experimental run.
- To empirically validate the privacy guarantees through shadow-model membership inference attack testing conducted on the final synthetic dataset, with a target attack success rate at or below 55%.
- To quantify the privacy-utility trade-off across at least three epsilon configurations, reporting epsilon values, membership inference attack success rates, and synthetic data quality scores for each configuration.

3. METHODOLOGY

3.1 Research Approach

This project follows a Design Science Research (DSR) methodology, which suits work that constructs a technical artifact, evaluates it against defined criteria, and draws knowledge from how it behaves under those criteria. The artifact is the ε-differential privacy enforcement component embedded in the AV synthetic data generation pipeline. DSR is combined here with a controlled experimental approach: multiple configurations of the privacy parameters will be run under identical conditions and compared quantitatively. This combination of artifact construction and systematic experimentation matches standard practice in applied privacy research, as seen in Abadi et al. [2] and Yousefpour et al. [3], where tools are built and then evaluated through controlled training runs. The independent variable across experiments is the privacy configuration, specifically the clipping norm C and noise multiplier sigma. The dependent variables are the cumulative ε value, the membership inference attack success rate, and the AV perception model Mean Average Precision (mAP) score. All experiments will cover at least three configurations so that results can be compared rather than taken from a single run.

All experiments will be run quantitatively. Privacy will be measured through the formal ε values from the RDP accountant and the empirical membership inference attack success rates. Utility will be assessed using the mAP scores reported by IT22110220. The key tools used across the methodology are Python 3.10 as the implementation language, PyTorch 2.x as the deep learning framework chosen for its dynamic computation graph and native Opacus integration, Opacus 1.x as the differential privacy library, and CARLA as the simulation environment managed by IT22541284. The Waymo Open Dataset is used only for calibration and benchmarking, not for model training, which keeps real-data exposure limited and within the terms of its research usage license.

3.2 Selecting the Privacy Mechanism and Library

The first step is choosing which differential privacy mechanism and software library to use. The Gaussian mechanism is the standard choice for deep learning because it works with stochastic gradient descent and has well-understood composition properties under the RDP framework. The key parameters are the clipping norm C, which limits how much any single training sample can influence the gradient, and the noise multiplier sigma, which controls how much noise is added. Higher sigma gives stronger privacy but reduces gradient signal.

Opacus will be evaluated as the primary implementation tool. It integrates directly with PyTorch, handles per-sample gradient computation without architectural changes to the model, and includes a built-in RDP accountant. TensorFlow Privacy will be kept as a backup option in case the group's generative pipeline ends up being implemented in TensorFlow rather than PyTorch. The library decision will be made at the start of Phase 1 based on what the simulation and generative modelling components require.

The epsilon value will be treated as a parameter to explore rather than a fixed target. Initial experiments will test values between 1 and 10, which covers the range used in most published deep learning privacy research. The final choice will depend on what the membership inference attack results show and how much the noise affects the quality of the synthetic data for AV training.

3.3 Privacy-Aware Calibration

Using real driving data to calibrate the simulator is where the most sensitive information enters the pipeline. The approach taken here will restrict the use of real data strictly to aggregate statistics, not individual records. Traffic density will be estimated from vehicle count distributions, and pedestrian behavior will be characterized by using population-level trajectory statistics rather than tracking specific people.

Those aggregate statistics will themselves be computed under the Laplace mechanism before being used for calibration. The Laplace mechanism adds noise proportional to the sensitivity of the query divided by epsilon. For count and histogram queries, the sensitivity is well defined, and the noise addition is straightforward. This means that even the indirect channel through which real data influences the simulation is formally protected by a privacy guarantee.

Any real-world sensor data used during calibration will go through a feature abstraction step that extracts only high-level scene properties, such as overall illumination distributions, object class frequencies, and weather condition statistics. Pixel-level and point-level detail that could carry identifying information will be discarded before any computation is done. The abstraction procedure will be written into the project code so that it is repeatable and auditable.

3.4 Differentially Private Training of the Generative Model

The generative model, whether GAN-based, VAE-based, or a hybrid, will be trained using Opacus. At each training step, Opacus computes the gradient for each sample separately, clips it to a maximum L2 norm of C, averages the clipped gradients across the batch, and adds Gaussian noise scaled to C times sigma before the optimizer applies the update. Clipping limits how much any single record can pull the model weights in a particular direction, and the noise on top of that prevents an attacker from working backwards from the weights to infer which records were in the training set.

Three ablation runs will be completed using different (C, sigma) pairs before the full-scale generation run in Phase 3 starts. Each run will be compared on its cumulative ε, mAP score on the downstream AV model, and FID score of the synthetic sensor outputs. Whichever configuration keeps ε within budget while producing the least degradation in mAP will be carried forward into the mass generation phase.

3.5 Managing the Privacy Budget

Every training step consumes a portion of the total privacy budget. The RDP accountant in Opacus tracks this consumption precisely across the full training run. A maximum budget of $\varepsilon_{sax} = 10$ will be set before training starts. If the accountant reaches this limit during training, the training loop will stop automatically or reduce the per-batch contribution rather than exceeding the budget. This prevents the privacy guarantee from degrading during the large-scale generation run in Phase 3.

The accountant output will be logged at every training step and saved alongside the training loss and validation metrics. This creates a privacy cost curve for every experiment, making it possible to see exactly when and where the budget is being spent. If certain phases of training are consuming budget faster than expected, the parameters can be adjusted before the full run.

3.6 Validation Through Attack Testing

The formal ε values from the RDP accountant give a mathematical bound on privacy, but formal bounds need to be backed by practical tests. Three categories of metrics will be used to evaluate the system. The first is privacy metrics: the shadow-model membership inference attack success rate, where the target is 55% or below (close to random guessing at 50%), and a re-identification test success rate below 10% when attempting to match synthetic trajectories back to real calibration records. The second is utility metrics: the mAP of an AV object detection model trained on the synthetic data, where the target is within 15% of the baseline mAP achieved on real Waymo Open Dataset frames, and the Fréchet Inception Distance (FID) score between synthetic and real sensor frames, where a lower FID indicates better statistical similarity. The third is privacy budget metrics: the total cumulative ε value logged by the RDP accountant, which must remain at or below $\varepsilon = 10$ across calibration and training combined. These three categories together cover the full privacy-utility trade-off that this project is designed to quantify.

Re-identification risk will be assessed by attempting to match synthetic driving trajectories back to the real calibration records using cross-referencing techniques. Results from every experimental configuration will be logged in a standardized results table recording the clipping norm C, noise multiplier sigma, cumulative ε, MIA attack success rate, FID score, and mAP score, so the full privacy-utility trade-off is visible across all runs and configurations.

All attack scripts will be developed in coordination with IT22066916 and stored in a shared, version-controlled repository. Data used in this project falls into two categories. The real-world calibration data consists of selected subsets from the Waymo Open Dataset, which contains 1,950 driving segments each 20 seconds long, captured with five cameras and five LiDAR sensors across a range of weather conditions and geographic locations, accessed under the Waymo Dataset License Agreement for non-commercial research use. Only the aggregate statistical properties of these segments will be extracted for calibration purposes. No raw frames or point clouds will be stored after the calibration statistics are computed. The synthetic training data will be generated entirely within the CARLA simulation environment configured by IT22541284, producing camera, LiDAR, and radar sensor outputs with automatically generated ground-truth labels. No real individuals appear in the synthetic data, which removes the primary source of ethical risk during training.

3.7 Ethical Considerations

The Waymo Open Dataset is used strictly for calibration statistics, not for model training. It is accessed under the Waymo Dataset License Agreement, which permits non-commercial research use of this kind. The calibration process will operate on aggregate statistics only. Raw frames and point clouds will be processed in memory and will not be written to disk, retained beyond the calibration step, or shared outside the research group. No attempt will be made to identify individuals from the dataset. If the scope of real data use changes during the project, ethical clearance will be sought through SLIIT's institutional ethics review process before any additional data is accessed. The synthetic dataset generated through CARLA contains no real individuals and carries no personal data, which removes the primary ethical risk associated with AI training pipelines of this kind. Informed by the privacy-by-design principles embedded in the GDPR [12] and the Sri Lanka Personal Data Protection Act No. 9 of 2022, the project treats data minimization and purpose limitation as non-negotiable constraints on how real data is handled throughout.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

The privacy integration component is the part of the group system responsible for ensuring that no individual-level information from real driving data can be recovered from the synthetic AV dataset or the models used to generate it. It sits between two boundaries: on one side it receives raw calibration data from the Waymo Open Dataset, and on the other it feeds privatized configuration parameters to the CARLA simulator operated by IT22541284. It also interfaces with the AV model training pipeline run by IT22110220 (which receives the final synthetic dataset) and the attack testing framework run by IT22066916 (which validates the privacy guarantees). Figure 1 below shows the overall layout of the component and its connections to the rest of the group system.

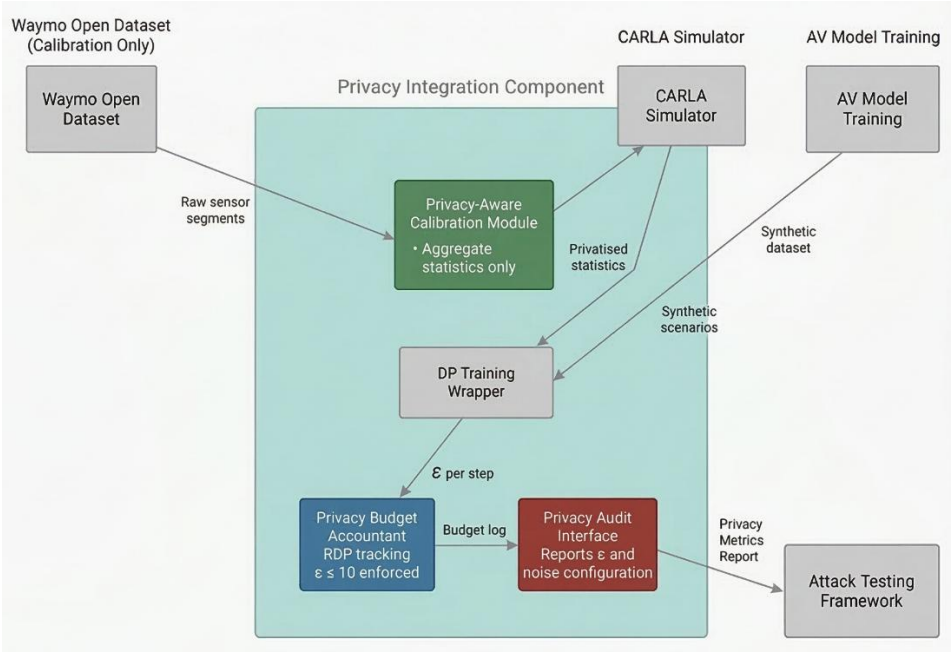


Figure 4-1: High-Level System Architecture

4.2 Sub-components

The privacy integration component (IT22309556) consists of four sub-modules, each with a distinct responsibility within the pipeline. The Privacy-Aware Calibration Module is the entry point for real-world data. It receives selected segments from the Waymo Open Dataset, extracts aggregate statistical properties such as vehicle count distributions and pedestrian trajectory patterns, applies the Laplace mechanism to add calibrated noise to each query result, and outputs only those privatized statistics to the simulator. Raw frames and point clouds are discarded in memory and never written to disk. The DP Training Wrapper wraps the generative model's PyTorch optimizer using Opacus. At every training step it computes per-sample gradients, clips each to the maximum L2 norm C, averages the clipped gradients, and adds Gaussian noise scaled to C times sigma before the optimizer applies the update. This is the core mechanism that enforces ε-DP during model training. The Privacy Budget Accountant runs alongside the DP Training Wrapper using the RDP composition framework built into Opacus. It accumulates the ε cost of each training step and halts training automatically if the total reaches the pre-set ceiling of ε = 10. It produces a full budget log at the end of every experimental run. The Privacy Audit Interface packages the final ε value, the noise

configuration (C and sigma), and the budget log into a structured report. That report is passed to IT22066916's attack testing framework so that the membership inference attack runs are conducted against a fully documented and reproducible privacy setup.

4.3 Data Flow

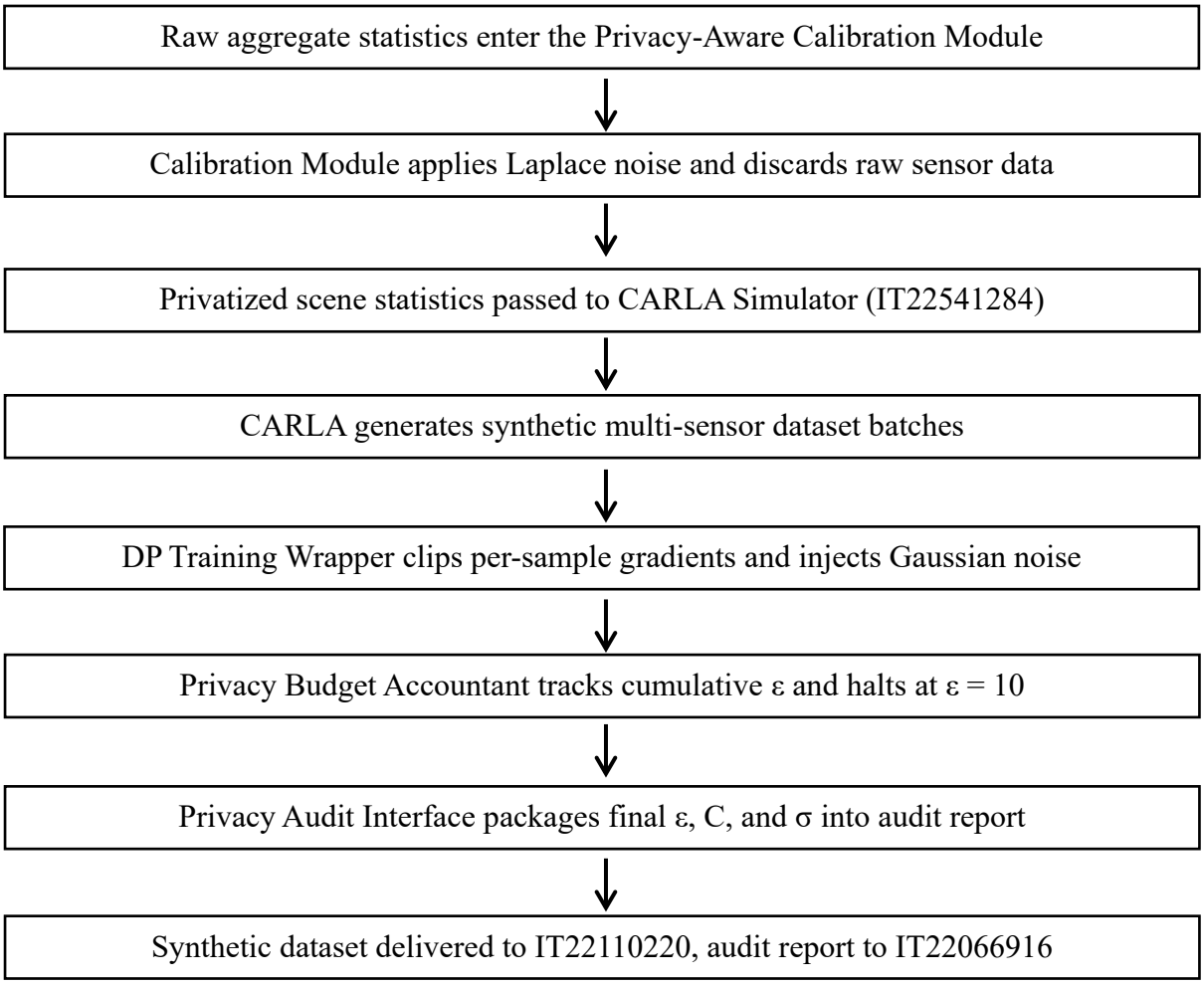


Figure 4-2: Data Flow Diagram

4.4 Security and Scalability

On the scalability side, per-sample gradient clipping is parallelizable across all samples in a batch, so adding more training data does not change the per-step cost of privacy enforcement in a way that creates a bottleneck. The RDP accountant adds negligible overhead as it performs scalar arithmetic alongside each training step. The main scalability constraint is GPU memory: Opacus requires storing per-sample gradients before clipping, which increases memory use compared to standard batch gradient computation. For large synthetic datasets this will be managed by reducing batch size or using gradient checkpointing, both of which are supported natively in the Opacus and PyTorch stack. On the security side, the threat model for this component covers three attack surfaces. The first is the calibration channel, where an adversary with access to the privatized statistics tries to recover individual Waymo records. The Laplace mechanism provides a formal ε-DP bound on this channel, meaning the adversary's ability to distinguish any individual record is bounded mathematically regardless of computational resources. The second is the model weights, where a membership inference attacker tries to

determine which records were in the training set by probing the generative model. The ε-DP guarantee from the DP Training Wrapper bounds this channel as well, and this is validated empirically through IT22066916's attack tests. The third is the synthetic dataset itself, where a re-identification attacker tries to link synthetic trajectories to real Waymo identities. Privacy-by-design in the calibration step, combined with the noise applied during generation, reduces the statistical similarity between synthetic outputs and specific real records. No raw Waymo data is ever stored to disk within this component, and all intermediate outputs are retained only for the duration of the computation that requires them.

5. USER REQUIREMENTS

5.1 Stakeholders

Primary stakeholders interact directly with the output of this component. AV development teams and research organizations need synthetic training data with an auditable privacy guarantee so they can use and share it without legal exposure under the GDPR or equivalent regulations. The group members (IT22541284, IT22110220, IT22066916) depend on it directly because its outputs feed into their simulation, utility evaluation, and attack testing pipelines. Prof. Anuradha Jayakody, as academic supervisor, is responsible for approving the research methodology and results. On the secondary side, data protection regulators may need to verify what privacy guarantees a synthetic dataset carries before accepting it as evidence of compliance. The individuals whose driving data appears in the Waymo Open Dataset have an interest in ensuring the calibration step does not expose their personal information, even though aggregate statistics. Future researchers in AV privacy may build on or replicate this methodology.

5.2 Functional Requirements

1. FR-01: Every training run runs under ε-differential privacy, with the ε value set in the configuration file before training starts. At the end of each run, the RDP accountant log must show the configured ε was not exceeded.
2. FR-02: The training pipeline clips each sample's gradient to a maximum L2 norm C before averaging across the batch, with C set in the configuration file. Gradient norm logs from any run must show no per-sample value above C.
3. FR-03: Gaussian noise scaled to C times sigma is injected into the averaged clipped gradients at every step, before the optimizer updates the weights. A unit test covering the training wrapper checks that the noise standard deviation matches sigma for each step.
4. FR-04: The Renyi Differential Privacy accountant tracks cumulative ε throughout training and cuts off the training loop once the configured ceiling is reached. The budget log for every run must record a final ε at or below ε = 10.
5. FR-05: Before any Waymo calibration statistics reach the simulator, Laplace noise is applied to each aggregate query result. No raw frames or point clouds from Waymo may appear in any output file or simulator input.
6. FR-06: At the close of every generation run, the system writes an audit report containing the final ε, clipping norm C, noise multiplier sigma, and total steps completed with no manual action required. The report must be present and fully populated for every run in the experiment log.
7. FR-07: Per-step ε cost and cumulative ε are written to a log at every training step and exported as a CSV. When loaded into Python pandas, the file must contain one row per step with columns for step number, per-step ε, and cumulative ε.

5.3 Non-Functional Requirements

- Privacy: Every training run must stay within the ε-DP bound set in the configuration file, confirmed by the RDP accountant log. Membership inference attack success rates against the resulting synthetic data must not exceed 55 percent under the shadow-model protocol run by IT22066916.

- Performance: The overhead from differential privacy enforcement must not increase total training time by more than 30 percent compared to a non-private baseline run using the same model architecture, dataset size, and hardware. GPU memory usage from per-sample gradient storage must not exceed twice the baseline memory footprint at the same batch size.
- Usability: The ε value, clipping norm C, and noise multiplier sigma must all be configurable through a single settings file, with no changes required to the generative model code. Switching to a different privacy configuration must require only editing that file and rerunning the training script.
- Reliability: The RDP accountant must halt training before the configured ε ceiling is breached in every run, including runs that encounter unexpected interruptions or restarts. Budget logs must be written atomically so that a crashed run cannot produce a partially written or corrupted log file.
- Security: Raw calibration data must never be written to disk within this component. All intermediate outputs must be held in memory only for the duration of the computation that requires them. Model weights and budget logs must be stored under file-system access controls restricting read and write permissions to the research group. Because ε-DP is enforced throughout training, an attacker probing either the model weights or the synthetic dataset is bounded in what they can learn about any individual by the configured ε value.
- Maintainability: Each of the four sub-modules must be built as a separate, independently testable Python module with its own unit tests. Swapping out any one sub-module must not require code changes in the others beyond updating the interface call.
- Reproducibility: All random seeds, noise configurations, and ε values used in any experiment must be logged automatically alongside the results, so any run can be reproduced exactly from the settings file and log alone.
- Compatibility: The DP Training Wrapper must work with the PyTorch-based generative model built by IT22541284, and the Privacy Audit Interface output must be readable by IT22066916's attack testing framework without any format conversion.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

AV companies and automotive manufacturers subject to GDPR, CCPA, or equivalent regulations are the primary market. Their ML engineers and data infrastructure leads need to train perception models at scale but cannot license or store real driving footage without triggering lengthy compliance reviews. Smart city monitoring and insurance telematics form a secondary market where the same ε-DP calibration approach applies.

6.2 Value Proposition

Synthetic AV training data certified under ε-DP and validated against membership inference attacks, with no real-world data licensing or consent infrastructure required. For the buyer, legal review time drops, data retention liability is removed, and an auditable privacy certificate is ready to present to regulators.

6.3 Revenue Model

Under a Software-as-a-Service model, customers submit calibration configuration and receive a certified synthetic dataset with an attached ε-DP audit report. A B2B licensing model embeds the ε-DP integration library into an existing simulation platform such as CARLA or NVIDIA DRIVE Sim under an annual license agreement.

6.4 Cost Estimation

Development costs cover cloud compute for ε ablation runs, storage for synthetic dataset batches, and any tooling not available under open-source licenses. Since Opacus and TensorFlow Privacy are both Apache 2.0, software licensing adds no expense. The primary cost driver is GPU time, as per-sample gradient clipping under ε-DP increases training duration relative to a non-private baseline.

6.5 Pricing Strategy

Organizations that run dataset generation repeatedly can pay a recurring subscription, with fees scaled to compute volume and ε configuration. Teams with occasional need pay per certified run instead. Simulator vendors, such as those building on CARLA or NVIDIA DRIVE Sim, can license the ε-DP pipeline directly and ship it as part of their own platform under an annual agreement.

6.6 Competitive Advantage

No current synthetic AV data platform demonstrates ε-DP compliance across both the calibration and training steps. NVIDIA DRIVE Sim and Wayve's tools produce high-fidelity outputs but carry no formal privacy certificate. That gap is where this component sits.

6.7 Intellectual Property Considerations

The pipeline architecture combining RDP budget accounting with multi-modal AV simulator calibration under the Laplace mechanism is a candidate for a provisional patent. Copyright in the implementation code sits with SLIIT under the university's IP policy unless a commercialization agreement transfers it. An Apache 2.0 release is also viable, with revenue drawn from the SaaS tier and support contracts rather than the code. Unpublished benchmark results stay as trade secrets.

7. BUDGET AND JUSTIFICATION

Table 7-1: The Budget

Item	Estimated Cost (LKR)	Justification
Personal Laptop	LKR 0	Development and testing run on an existing personal laptop. No new hardware purchase is required. Compute-intensive training runs go to cloud GPU instances rather than a dedicated local GPU.
Cloud GPU Compute (AWS, approx. 200 hours)	LKR 32,000	DP training runs slower than standard training due to per-sample gradient computation. Multiple ablation experiments across different ε settings must be completed before the final generation run.
Cloud Storage (1 TB)	LKR 8,000	Storage for synthetic dataset batches, model checkpoints, and privacy budget logs across all experimental runs.
Opacus and TensorFlow Privacy (open source)	LKR 0	Both libraries are freely available under open-source licenses and do not require any paid subscription or seat license.
Research paper access (IEEE Xplore, ACM DL)	LKR 8,000	Key papers on differentially private GANs and AV data privacy sit behind paywalls and require institutional or individual purchase.
Human effort (research and implementation)	LKR 0	The academic project scope covers all research and implementation work. No external or commercial labor cost applies.
Total	**LKR 48,000**	

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Tasks are assigned to members according to their individual research focus. Phases, weeks, and key tasks for all four members are listed on the table.

Table 8-1: Task Breakdown per Member

Member	Role	Phase	Weeks	Key Tasks
IT22309556	Privacy Integration	1	1-3	Literature review on DP mechanisms. Library selection (Opacus vs TF Privacy). Define ε range for ablation.
IT22309556	Privacy Integration	2	4-7	Build DP training wrapper. Implement Privacy-Aware Calibration Module with Laplace noise. Run ablation experiments.
IT22309556	Privacy Integration	3	8-10	Full-scale generation run under ε-DP. Monitor ε consumption. Confirm budget not exceeded.
IT22309556	Privacy Integration	4	11-14	Coordinate with IT22066916 on MIA tests.
IT22541284	Dataset Generation / Simulation	1	1-3	Set up CARLA simulator environment. Define scene and sensor configurations.
IT22541284	Dataset Generation / Simulation	2	4-8	Run CARLA simulation using privatized calibration statistics from IT22309556. Generate raw synthetic dataset batches.
IT22541284	Dataset Generation / Simulation	3	9-12	Validate dataset format and sensor coverage. Deliver final synthetic dataset to IT22110220.
IT22110220	Utility Evaluation	1	1-4	Define utility metrics (mAP, FID). Set up AV model training pipeline and baseline benchmarks.
IT22110220	Utility Evaluation	2	8-11	Train AV model on synthetic dataset. Measure mAP and FID against non-private baseline.
IT22110220	Utility Evaluation	3	12-14	Report utility trade-off results. Contribute utility analysis to group thesis.
IT22066916	Risk and Compliance	1	1-3	Review GDPR, CCPA, and Sri Lanka PDPA requirements. Define compliance checklist.
IT22066916	Risk and Compliance	2	9-12	Run membership inference and re-identification attack tests on DP synthetic data from IT22309556.
IT22066916	Risk and Compliance	3	12-14	Compile attack results and compliance report. Contribute risk and regulatory sections to group thesis.

8.2 Timeline Alignment

Shaded cells mark weeks where a member has active tasks. Phases overlap at inter-member handoff points: IT22309556 delivers privatized calibration statistics to IT22541284 at the end of Phase 1 and delivers the ε-DP audit log to IT22066916 at the start of Phase 3.

Table 8-2: Timeline Alignment

Member	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
IT22309556	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IT22541284	1	2	3	4	5	6	7	8	9	10	11	12		
IT22110220	1	2	3	4				8	9	10	11	12	13	14
IT22066916	1	2	3						9	10	11	12	13	14

8.3 Balanced Workload Distribution

Active week counts run from 9 to 14 across the four members. IT22309556 covers all 14 weeks because the ε budget requires continuous monitoring throughout every phase. IT22541284 and IT22110220 each run 11 to 12 active weeks. IT22066916 covers 9 weeks, concentrated in the attack testing and compliance reporting phase. The difference in week counts follows from the nature of each role: privacy integration runs end-to-end, while utility evaluation and compliance testing fall within bounded phases.

Table 8-3: Balanced Workload Distribution

Member	Active Weeks	Phases	Primary Deliverables
IT22309556	1-14 (14 weeks)	4	ε-DP training wrapper, calibration module, ablation results, privacy budget log
IT22541284	1-12 (12 weeks)	4	CARLA simulation setup, synthetic dataset batches, validated dataset delivery
IT22110220	1-4, 8-14 (11 weeks)	4	Utility metric framework, AV model training results, mAP and FID analysis report
IT22066916	1-3, 9-14 (9 weeks)	4	Compliance checklist, MIA and re-identification attack results, regulatory compliance report

9. GANTT CHART

Table 9-1: Gantt Chart

Task	Semester 1 (Mar – May 2026)			Semester 2 (Jun – Dec 2026)						
All Members	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Literature Deep Dive and Architecture Mapping										
Hardware and Storage Allocation Setup										
Calibration Strategy and Integration Checkpoint										
Full Pipeline Execution and QA Review										
Metric Compilation and Final Paper Drafting										
SLIIT ASSESSMENT MILESTONES										
★ Proposal Submission (15 Mar 2026)	★									
★ Proposal Presentation (16–18 Mar 2026)	★									
★ Progress Presentation 1 (4–6 May 2026)			★							
★ Progress Presentation 2 (17–19 Aug 2026)						★				
★ Final Report Submission (11 Oct 2026)								★		
★ Final Presentation and Viva (12–14 Oct 2026)								★		
★ Research Paper Submission (15 Oct 2026)								★		
★ Publication Evidence Submission (1 Dec 2026)										★

Star cells mark SLIIT official milestones. Semester 1 runs from March to May 2026; Semester 2 runs from June to December 2026.

10. REFERENCES LIST

[1] C. Dwork, "Differential Privacy," in Proc. 33rd Int. Colloq. Automata, Languages and Programming (ICALP), Springer, 2006, pp. 1-12.

[2] M. Abadi et al., "Deep Learning with Differential Privacy," in Proc. 2016 ACM SIGSAC Conf. Computer and Communications Security (CCS), 2016, pp. 308-318.

[3] A. Yousefpour et al., "Opacus: User-Friendly Differential Privacy Library in PyTorch," arXiv preprint arXiv:2109.12298, 2021.

[4] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An Open Urban Driving Simulator," in Proc. 1st Annual Conf. Robot Learning (CoRL), 2017, pp. 1-16.

[5] S. Shah, D. Dey, C. Lovett, and A. Kapoor, "AirSim: High-Fidelity Visual and Physical Simulation for Autonomous Vehicles," in Field and Service Robotics, Springer, 2018, pp. 621-635.

[6] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership Inference Attacks against Machine Learning Models," in Proc. IEEE Symp. Security and Privacy (S&P), 2017, pp. 3-18.

[7] N. Carlini et al., "Extracting Training Data from Large Language Models," in Proc. USENIX Security Symp., 2021, pp. 2633-2650.

[8] A. Narayanan and V. Shmatikov, "Robust De-anonymization of Large Sparse Datasets," in Proc. IEEE Symp. Security and Privacy (S&P), 2008, pp. 111-125.

[9] L. Xie, K. Lin, S. Wang, F. Wang, and J. Zhou, "Differentially Private Generative Adversarial Network," arXiv preprint arXiv:1802.06739, 2018.

[10] R. Torkzadehmahani, P. Kairouz, and B. Paten, "DP-CGAN: Differentially Private Synthetic Data and Label Generation," in Proc. IEEE/CVF Conf. CVPR Workshops, 2019.

[11] I. Goodfellow et al., "Generative Adversarial Networks," in Advances in Neural Information Processing Systems (NeurIPS), 2014, pp. 2672-2680.

[12] European Union, "General Data Protection Regulation (GDPR)," Regulation (EU) 2016/679, Official Journal of the European Union, 2016.

11.APPENDICES

Appendix A: Differential Privacy Mechanism Summary

Three mechanisms handle privacy enforcement across the pipeline. Each is mapped to its role, noise type, and supporting library.

Mechanism	Where Used	Noise Type	Library
Gaussian Mechanism	Gradient noise during generative model training	Gaussian, noise multiplier sigma	Opacus (primary), TF Privacy (backup)
Laplace Mechanism	Aggregate statistics at calibration step	Laplace, scale = sensitivity / ε	Custom implementation using NumPy
RDP Accountant	Privacy budget tracking across training run	Accounting only, no noise added	Built into Opacus

Appendix A: Differential Privacy Mechanism Summary

Appendix B: Risk Analysis

Four risks specific to ε-DP integration are identified below, each with likelihood, impact, and mitigation.

Risk	Likelihood	Impact	Mitigation
Privacy budget runs out before training is complete	Medium	High	Pre-calculate expected ε consumption per phase before the full run. Use subsampling amplification to reduce the per-step budget cost.
Noise from ε-DP degrades synthetic data quality below useful threshold	Medium	High	Run ablation study across multiple (C, sigma) configurations before committing to the full generation run.
Opacus is not compatible with the generative model architecture chosen by the group	Low	Medium	TensorFlow Privacy is available as a backup. Both libraries are actively maintained and cover standard GAN and VAE architectures.
Membership inference attack results are ambiguous and difficult to interpret	Low	Medium	Use multiple attack variants including shadow model and likelihood ratio attacks. Compare results against the formal ε bound from the accountant.

Appendix B: Risk Analysis

Appendix C: Detailed Diagrams

Each component in the system architecture (Figure 1) is described below, covering its inputs, outputs, and data flow direction.

Component	Description
Privacy-Aware Calibration Module	Receives raw Waymo Open Dataset aggregate statistics. Applies Laplace noise scaled to sensitivity / ε before passing privatized statistics to the CARLA simulator. Raw frames and point clouds are discarded in memory and never written to disk.
DP Training Wrapper	Wraps the PyTorch optimizer using Opacus. Clips per-sample gradients to L2 norm C, averages across the batch, and injects Gaussian noise scaled to C x sigma before the weight update.
Privacy Budget Accountant	RDP accountant built into Opacus. Tracks cumulative ε after every training step. Halts the training loop automatically when cumulative ε reaches the configured ceiling (default: 10).
Privacy Audit Interface	Packages the final ε value, noise configuration (C, sigma), and total training steps into a structured report delivered to IT22066916 for membership inference and re-identification testing.
Data Flow (calibration channel)	Waymo aggregate stats -> Laplace noise injection -> CARLA simulator -> synthetic dataset batches.
Data Flow (training channel)	Synthetic dataset -> DP Training Wrapper -> model weights -> AV model training (IT22110220).
Data Flow (audit channel)	ε per step -> RDP Accountant -> cumulative ε log -> Privacy Audit Interface -> IT22066916.

Appendix C: Detailed Diagram

Appendix D: Ethical Documents

Ethical obligations, license agreements, and regulatory requirements governing this project are listed below.

Document / Obligation	Detail
Waymo Open Dataset License Agreement	Dataset used under the Waymo Dataset License Agreement for non-commercial academic research. Raw data processed in-memory only. No raw frames or point clouds written to disk or shared externally.
GDPR Article 25 (Data Protection by Design)	Privacy-by-design enforced through ε-DP at the calibration and training steps. Data minimization applied: only aggregate statistics from real data enter the pipeline.
Sri Lanka Personal Data Protection Act No. 9 of 2022	Purpose limitation and data minimization principles observed. No personal data stored beyond the duration of in-memory processing.

Appendix D: Ethical Documents